

RTD IN LFR

Name: Purva Kaushalbhai Shah

Group members name:

Ashwini Bukke

Mukul Meena

Pavani Khale

Naim Khan

Table of Contents:

Objective

Theory and Background

Known parameters

Formulae

Readings

Results and Discussion

Conclusion

References

Objective:

To determine the mean residence time distribution, to plot exit age distribution curve in the laminar flow reactor, to compare experimental and theoretical E curves.

Theory and Background:

A **Laminar Flow Reactor (LFR)** is a type of chemical reactor where the fluid flows in a laminar regime, meaning the flow is smooth and characterized by parallel layers with no disruption between them. This type of flow occurs at low Reynolds numbers, typically less than 2,100. The Residence Time Distribution (RTD) is a crucial tool for analyzing the flow behavior of reactors, especially non-ideal reactors like the LFR, where deviations from the plug flow model occur.

Residence Time Distribution (RTD) of a chemical reactor is a probability distribution function that describes the amount of time fluid elements spend inside a reactor. It characterizes the flow and mixing behavior in chemical reactors.

The time distributions of different routes that fluid elements take through the reactor may be different due to the fact that different portions of the fluid are in contact with different elements or wall zones of the reactor. The exit age distribution, or residence time distribution, is denoted by E. E is measured in unit of time^{-1} . The convenient way to represent RTD is in such a way that the area under the curve equals one, i.e. $\int_0^{\infty} E(t) dt = 1$. The value $E(t)dt$ is the fraction of the fluid that spends time duration t inside the reactor.

The RTD function E can be found using the equation : $E(t) = \frac{C(t)}{\int_0^{\infty} C(t) dt}$.

RTD is used to get the *Mean residence time* (t_m) and the variance.

Mean residence time (t_m) is the average time fluid elements spend in the reactor and variance is a measure of dispersion around the mean residence time. σ^2 (*variance*) and (t_m) help diagnose the performance of real reactors compared to ideal models, assess mixing quality, and predict the reactor's effectiveness in achieving desired conversions.

The integration of all the variables is done using the area under the curve of respective graphs between the respective variables.

The area is calculated using the shoelace formula. The **shoelace formula**, also known as the **surveyors' formula**, is used to find the area of a polygon given its vertices. In our case (of

graph), we are considering the area under the curve a polygon. To complete the polygon, in every graph I have added a point (end time, 0).

Shoelace formula:

For a polygon with vertices $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the area is given by:

$$Area = \frac{1}{2} \cdot \left| \sum_{i=1}^n (x_i y_{i+1} - y_i x_{i+1}) \right|$$

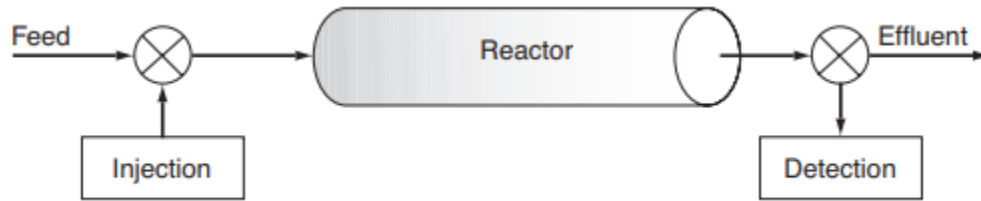


Figure 16-3 Experimental set up to determine $E(t)$.

Figure 0: Schematics of the experiment [1]

Known parameters:

- Volume of the reactor (V) = 1400 mL
- Volumetric flow rate (v) = $1 \left(\frac{\text{mL}}{\text{s}} \right)$
- Temperature = 27°C

Observation table (mentioned in manual)

Parameter	Value	unit
Temperature, T	27.0000	$^\circ\text{C}$
Reactor volume, V	1400.0000	mL
Volumetric flow rate, v	1.0000	ml/s
Space time (τ)	1400.0000	s

We are using 10g NaOH in 10 mL water, i.e **25 mM** solution for pulse input.

Formulae:

- Space time, $\tau = \frac{V}{v} = \frac{1400 \text{ mL}}{1 \frac{\text{mL}}{\text{s}}} = 1400 \text{ s}$
 - Where V is the Volume of the reactor
 - v is the Volumetric flow rate
- Exit age distribution (experimental): $E(t) = \frac{C(t)}{\int_0^\infty C(t) dt}$

- Where $C(t)$ is the Concentration
- t is the time
- Exit age distribution (theoretical) : $E(t) = \frac{4t}{t_m} e^{\frac{-2t}{t_m}}$
- Mean residence time $t_m = \int_0^{\infty} t * E(t) dt$
- Variance (σ^2) = $\int_0^{\infty} (t - t_m)^2 E(t) dt$
- Standard deviation (σ) = $\sqrt{\text{Variance}}$

Readings:

TABLE 1: TABLE CONTAINING EXPERIMENTAL DATA OF CONCENTRATION AND CONDUCTIVITY

Concentration (N)	Conductivity (mS)
0.05	10.8600
0.025	5.0460
0.0125	2.0800
0.00625	1.2910

TABLE 2: TABLE CONTAINING EXPERIMENTAL DATA OF CONDUCTIVITY AND TIME

Time (min)	Conductivity (mS)
3	386.4
6	390.3
9	442.9
12	439.8
15	412
18	408.4
21	405.1
24	400.4
27	395.4
30	389.8

TABLE 3: TABLE CONTAINING CALCULATED DATA

Conductivity (mS)	Time (min)	Concentration (N)	E (t)	t * E(t)	$(t - t_m)^2$	$(t - t_m)^2 E(t)$
386.4	3	1.7215	0.0302	0.0906	200.5938	6.0582
390.3	6	1.7388	0.0305	0.1830	124.6151	3.8015
442.9	9	1.9729	0.0346	0.3115	66.6364	2.3064
439.8	12	1.9591	0.0344	0.4124	26.6578	0.9162
412	15	1.8354	0.0322	0.4830	4.6791	0.1507
408.4	18	1.8194	0.0319	0.5745	0.7004	0.0224
405.1	21	1.8047	0.0317	0.6649	14.7217	0.4661
400.4	24	1.7838	0.0313	0.7511	46.7430	1.4628
395.4	27	1.7615	0.0309	0.8344	96.7643	2.9904
389.8	30	1.7366	0.0305	0.9140	164.7856	5.0205
388.9	33	1.7326	0.0304	1.0031	250.8069	7.6237

TABLE 4: TABLE CONTAINING E(t)

E(t) exp	E(t) theo
0.0302	0.0287273
0.0305	0.0405018
0.0346	0.0428267
0.0344	0.0402533
0.0322	0.0354700
0.0319	0.0300048
0.0317	0.0246766
0.0313	0.0198805
0.0309	0.0157662
0.0305	0.0123491
0.0304	0.0095758

Results and discussion:

Concentration (N) vs. conductivity (mS)

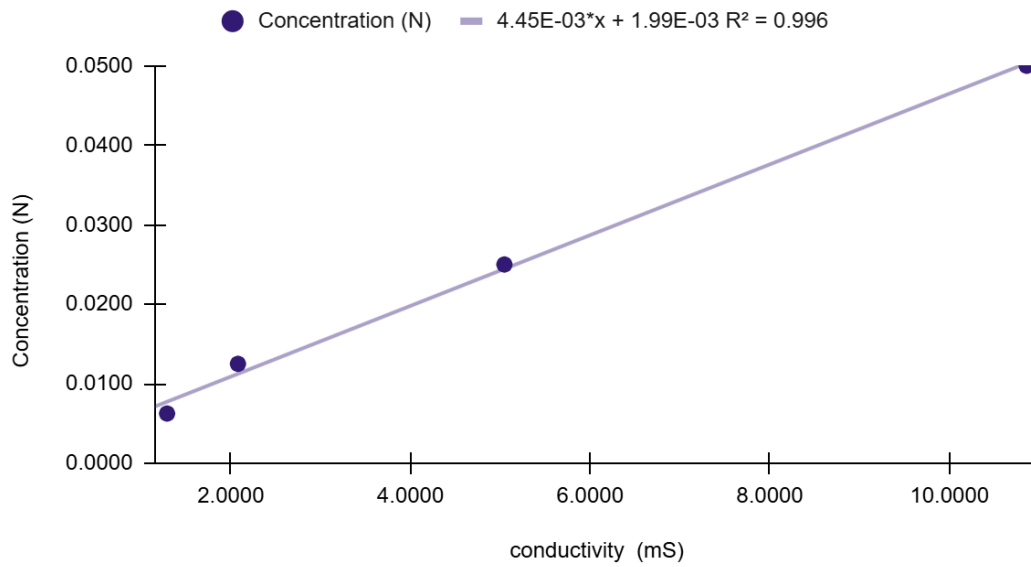


Figure 1: Figure showing the between Concentration (N) vs Conductivity (mS)

Concentration (N) vs time (min)

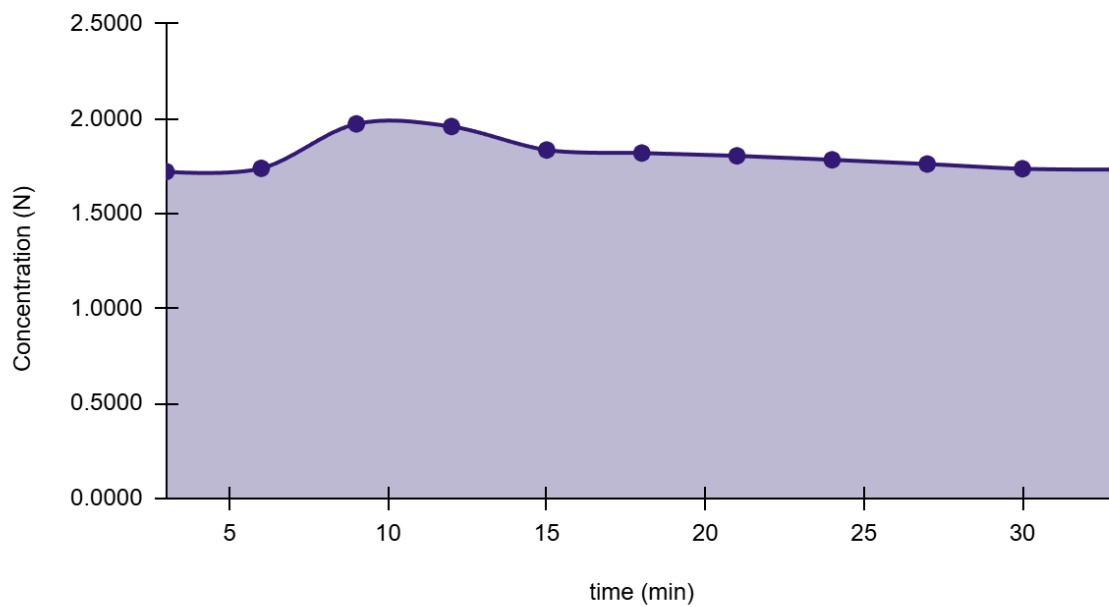


Figure 2: Figure showing the plot between Concentration (N) vs Time (min)

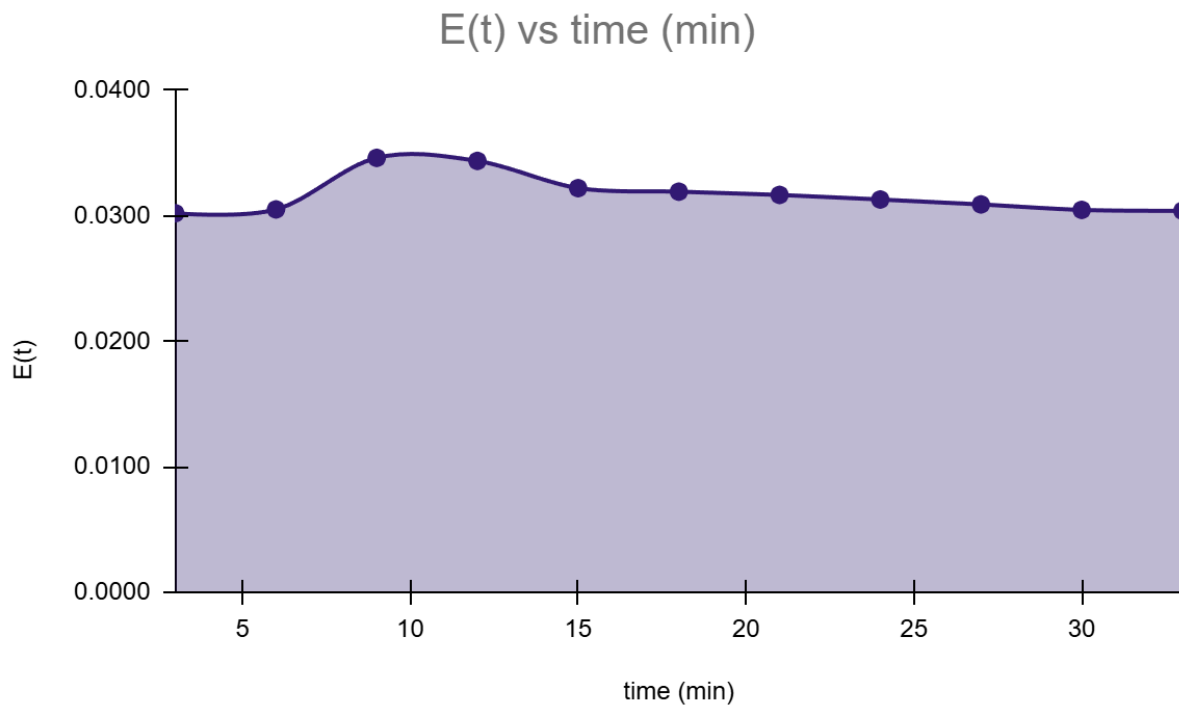


Figure 3: Figure showing the plot between $E(t)$ vs Time (min)

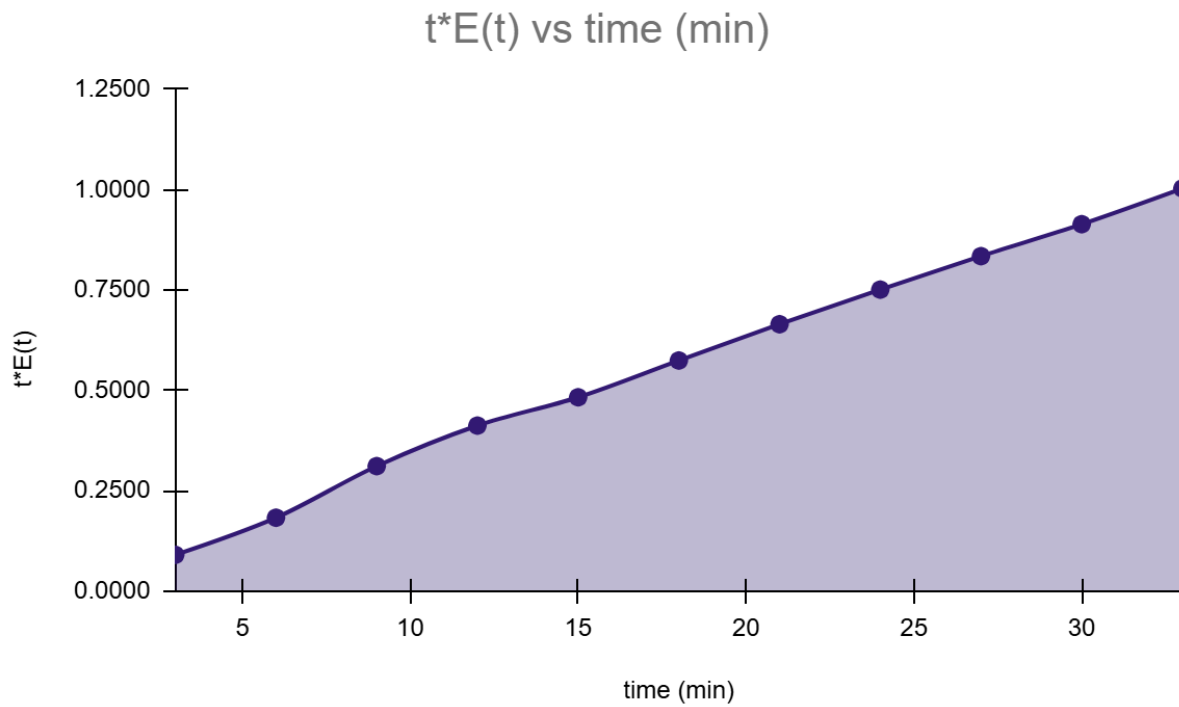


Figure 4: Figure showing the plot between $t \cdot E(t)$ vs Time (min)

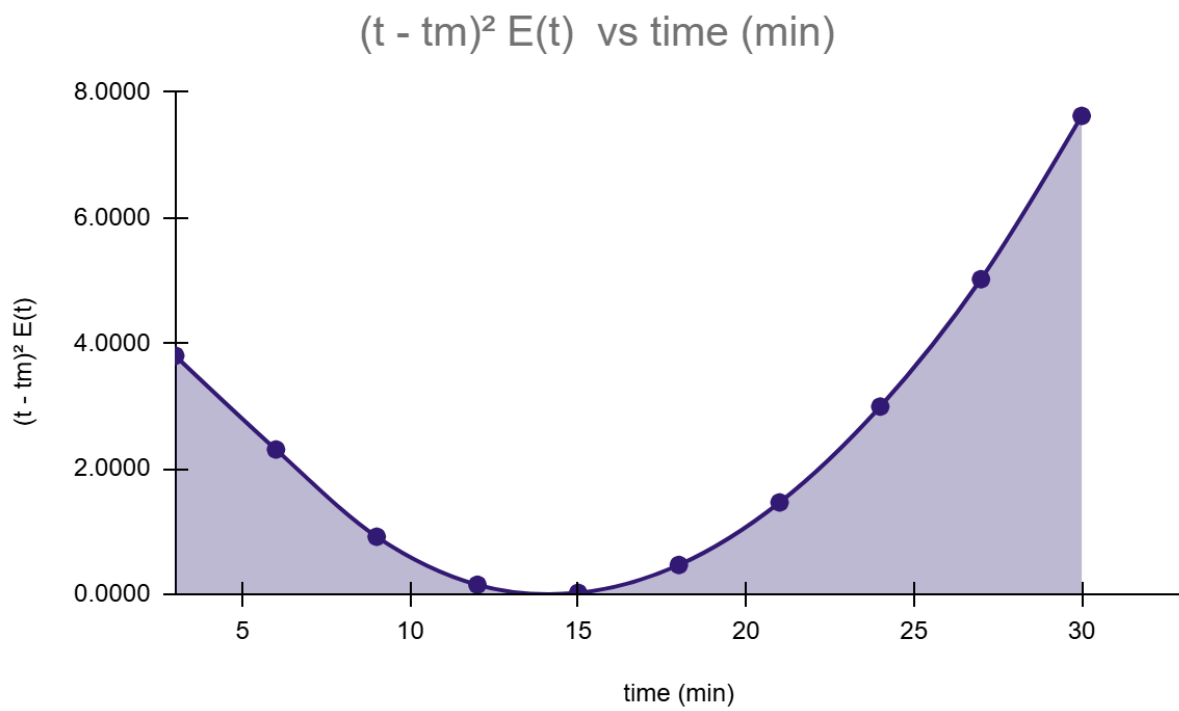


Figure 5: Figure showing the plot between $(t - t_m)^2 E(t)$ vs Time (min)

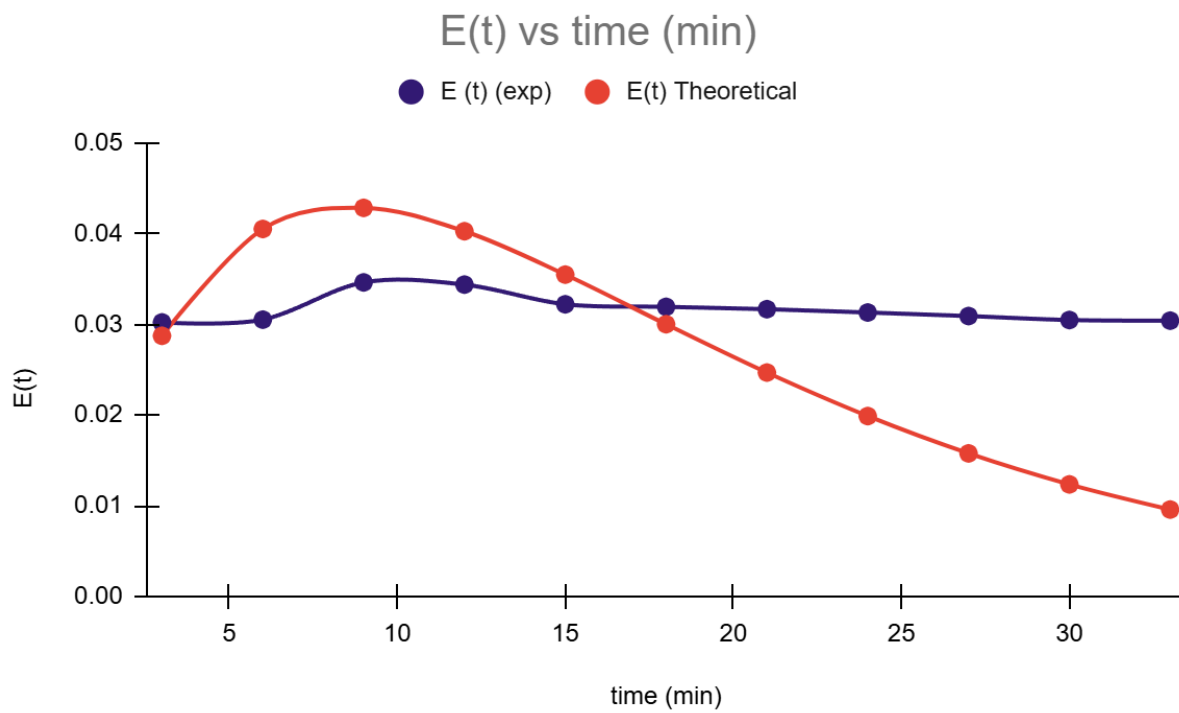


Figure 6: Figure showing the plot between $E(t)$ theoretical, $E(t)$ experimental vs Time (min)

The mean residence time for the LFR is = 17.16 min
Standard deviation for the LFR is = 9.0012

From figure 1, we can see that there is a linear relationship present between Concentration and conductivity, i.e more is the concentration, more is the conductivity.

Figure 2 shows the relationship between concentration and time. We can see that there is an increase in the concentration initially as we have injected the NaOH (pulse input), the NaOH starts mixing with the reactor contents. Due to the flow and mixing happening in the reactor, with time, the NaOH concentration at the exit is bound to rise. The concentration then becomes a maximum when most of the NaOH injected starts exiting the reactor. This equals the mean residence time (t_m) of the average fluid elements that spent typical amounts of time in the reactor. After maximum concentration, the NaOH concentration decreases, as most of the NaOH has already exited, and the remaining portions undergo dilution due to mixing and less residual NaOH amounts.

Figure 3 shows the relationship between E(t) (Exit age distribution) and Time and X*E(t) vs time respectively. The nature of the curve is the same as the concentration-time curve. E(t) is a probability density function derived from the concentration data. It is normalized so that the area under the curve equals 1. Therefore, the E(t) vs. time has the same shape as the concentration vs. time curve, but its scale is different due to normalization

Figure 4 shows the $t \cdot E(t)$ vs time curve and it is an increasing curve. The curve represents the weighted residence time distribution, which highlights the contribution of different residence times to the overall mean residence time (t_m).

Figure 5 shows the $(t - t_m)^2 E(t)$ vs Time curve. The curve first decreases as we approach towards t_m and after than the curve increases as we move far from t_m .

Figure 6 shows the theoretical E(t) vs time curve. This curve also first increases and then decreases.

The final results:

Mean residence time (t_m)	17.16 min
Variance (σ^2)	81.0210
Standard deviation (σ)	9.0012

Conclusion:

The study successfully determined the mean Residence Time Distribution (RTD) for a Laminar Flow Reactor (LFR), revealing critical insights into the reactor's flow and mixing behavior. By introducing a pulse input of 25 milliMolar NaOH and analyzing the concentration at the outlet, the exit age distribution $E(t)$ was obtained. The mean residence time (t_m) was 17.16 minutes, along with a variance of 81.0210 and a standard deviation of 9.0012, demonstrated the extent of residence time variations within the reactor.

This comparison between the experimental and theoretical values of the exit age distribution $E(t)$ shows significant discrepancies in their respective curves. The experimental values indicate that a larger proportion of fluid elements leave the reactor earlier, while the theoretical values from an idealized model show a slower distribution. The differences may arise from a variety of causes that include non-ideal flow patterns, imperfections in reactor design, or variations in fluid properties that are neglected in the theoretical model.

The results highlight that, in an LFR, fluid particles experience non-uniform residence times due to the parabolic velocity profile inherent to laminar flow. Some fluid elements exit quickly, while others remain in the reactor longer, resulting in a broad $E(t)$ curve. This distribution emphasizes the departure from ideal plug flow behavior, where all particles would have the same residence time.

The findings underscore the critical role of RTD in evaluating and optimizing reactor performance. Variations in residence time significantly impact reaction outcomes by affecting mixing efficiency and conversion. In the context of LFRs, understanding RTD is vital for predicting reactor behavior, designing effective processes, and improving operational performance.

These experimental data and their analysis provide additional support for the assertion that the correct treatment of the RTD is essential in chemical reactor design, especially in cases where non-ideal flow patterns exist. With the completion of this experiment, I have gained significant insights into how RTD plays an important role in reactor performance.

Sources of errors:

- While measuring the conductivity there are charges that we may have previous sample liquid in the bulb which is used to measure the conductivity.
- The tip to the conductivity measuring device may have touched the bottom/walls of the beaker.

References:

[1]“6.1 General Considerations.” Available: [Link](#)

[2] ‘Lab Manual’ provided in the classroom.

[3]“Art of Problem Solving,” *Artofproblemsolving.com*, 2025. [Shoelace Theorem](#)